

Analytical verification for transient uniaxial strain stiffening

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1 Problem

We solve

$$\nabla \cdot ((1 + \alpha \sqrt{\varepsilon \cdot \varepsilon}) \sigma) = 0, \quad (1)$$

where α is the `scalingFactor` ($\alpha = 0$ for no strain stiffening), on the following domain and boundary conditions:

- 1×1 m square
- `left_edge`: fixed ($u_x = u_y = 0$)
- `right_edge`: fixed in y-direction ($u_y = 0$), with forcing load F .

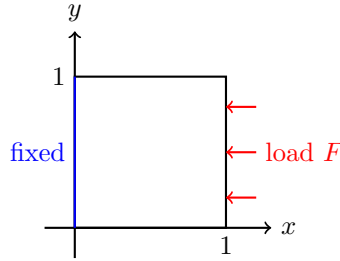


Figure 1: Geometry and boundaries

2 Analytical solution

Reducing the full tensor stress-strain relation $\sigma = C\varepsilon$ to two dimensions with uniaxial load ($\sigma_{xy} = \sigma_{yy} = 0$) and uniaxial deformation ($u_y = 0 \rightarrow \varepsilon_{xy} = \varepsilon_{yy} = 0$) leads to the following system of equations in Voigt notation

$$\begin{pmatrix} \sigma_{xx} \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2\mu + \lambda & \lambda & 0 \\ \lambda & 2\mu + \lambda & 0 \\ 0 & 0 & \mu \end{pmatrix} \begin{pmatrix} \varepsilon_{xx} \\ 0 \\ 0 \end{pmatrix}$$

and the relation

$$\sigma_{xx} = (2\mu + \lambda)\varepsilon_{xx}.$$

Plugging into equation 1

$$\frac{d}{dx}(1 + \alpha|\varepsilon_{xx}|)\sigma_{xx} \stackrel{!}{=} 0$$

leads to the quadratic equation (negative F to fit with openCFS implementation of pressure value)

$$\begin{aligned} \alpha(2\mu + \lambda)|\varepsilon_{xx}|\varepsilon_{xx} + (2\mu + \lambda)\varepsilon_{xx} &= \frac{-F}{A} \\ \Leftrightarrow \alpha|\varepsilon_{xx}|\varepsilon_{xx} + \varepsilon_{xx} + \frac{F}{A(2\mu + \lambda)} &= 0 \end{aligned} \quad (2)$$

where A is the area of the applied force. We solve two cases:

1. **Compression** $\varepsilon_{xx} < 0$: When we have $\varepsilon_{xx} = \frac{\partial u_x}{\partial x} < 0$, the domain is compressed. Solving equation 2 with $|\varepsilon_{xx}| = -\varepsilon_{xx}$ leads to

$$\varepsilon_{xx} = \frac{1 - \sqrt{1 + (4\alpha F)/(A(2\mu + \lambda))}}{2\alpha},$$

which is negative if and only if $F > 0$ (except for a rare case of negative λ), so the solution makes sense.

2. **Tension** $\varepsilon_{xx} > 0$: When we have $\varepsilon_{xx} = \frac{\partial u_x}{\partial x} > 0$, the domain is stretched. Solving equation 2 with $|\varepsilon_{xx}| = \varepsilon_{xx}$ leads to

$$\varepsilon_{xx} = \frac{-1 + \sqrt{1 - (4\alpha F)/(A(2\mu + \lambda))}}{2\alpha}$$

which is positive if and only if $F < 0$ (except for a rare case of negative λ), so the solution makes sense.

Since $\varepsilon_{xx} = \partial u_x / \partial x$ and $u_x|_{x=0} = 0$ (because `left_edge` is fixed), we get

$$u_x(x) = \varepsilon_{xx}x.$$

3 Results comparison

The mesh consists of one quadrilateral element. Parameters:

- Scaling factor $\alpha = 2.0$
- Elasticity modulus $E = 200$
- Poisson ratio $\nu = 0.3$
- Force area $A = 1$

	Test 1	Test 2	Test 3	Test 4
Pressure	-200	200	400	600
Analytical	0.40873	-0.40873	-0.64741	-0.83479
Numerical	0.40873	-0.40873	-0.64741	-0.83479

Table 1: Comparison of maximum displacement $u_x(1)$ for different tests.